

Project Title: FlowGuard: Supply Chain Predictive Analytics

Problem Statement

Manual demand forecasting and fragmented data pipelines are causing significant inventory inefficiencies. Currently, the business struggles with stockouts on high-demand items and overstocking of slow-moving inventory, leading to inflated holding costs and lost revenue opportunities.

Proposed Solution

Implement "FlowGuard," an automated end-to-end data pipeline and predictive analytics model. The solution extracts raw inventory and sales data, processes it through automated quality gates using a Medallion architecture (Bronze, Silver, Gold layers), and feeds a predictive forecasting model. Output is delivered via a user-friendly automated dashboard, completely replacing manual Excel-based forecasting.

Financial Impact (ROI, Payback & Annual Savings)

- **Current Baseline:** Estimated annual inventory holding costs of Ksh 5,000,000.
- **Projected Savings:** A conservative 15% reduction in excess inventory holding costs = **Ksh 750,000 / year.**
- **Investment Cost:** Cloud infrastructure setup and 3 months of internal resource allocation = **Ksh 150,000.**
- **Math & Metrics:**
 - **Net Profit (Year 1):** Ksh 750,000 - Ksh 150,000 = Ksh 600,000
 - **ROI:** $(\text{Net Profit} / \text{Investment}) \times 100 \rightarrow (600,000 / 150,000) \times 100 = \mathbf{400\%}$
ROI
 - **Payback Period:** $\text{Investment} / (\text{Annual Savings} / 12) \rightarrow 150,000 / (750,000 / 12) = \mathbf{2.4 \text{ months}}$

Risks & Mitigation

1. **Risk:** Data quality issues from legacy entry systems skewing the forecast.
 - **Mitigation:** Implement strict automated data quality gates in the ETL workflow before data reaches the reporting layer.
2. **Risk:** Low adoption rate by the procurement team accustomed to legacy processes.
 - **Mitigation:** Roll out the dashboard in phases, providing hands-on training and retaining a familiar UI layout using Next.js/Tailwind CSS.

Recommendation GO.

We request immediate approval for the Ksh 150,000 deployment budget. The project pays for itself in less than 3 months and establishes a scalable foundation for future analytics initiatives.